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Blog

Study Finds U.S. Could Run Solely on Renewable Energy by 2050

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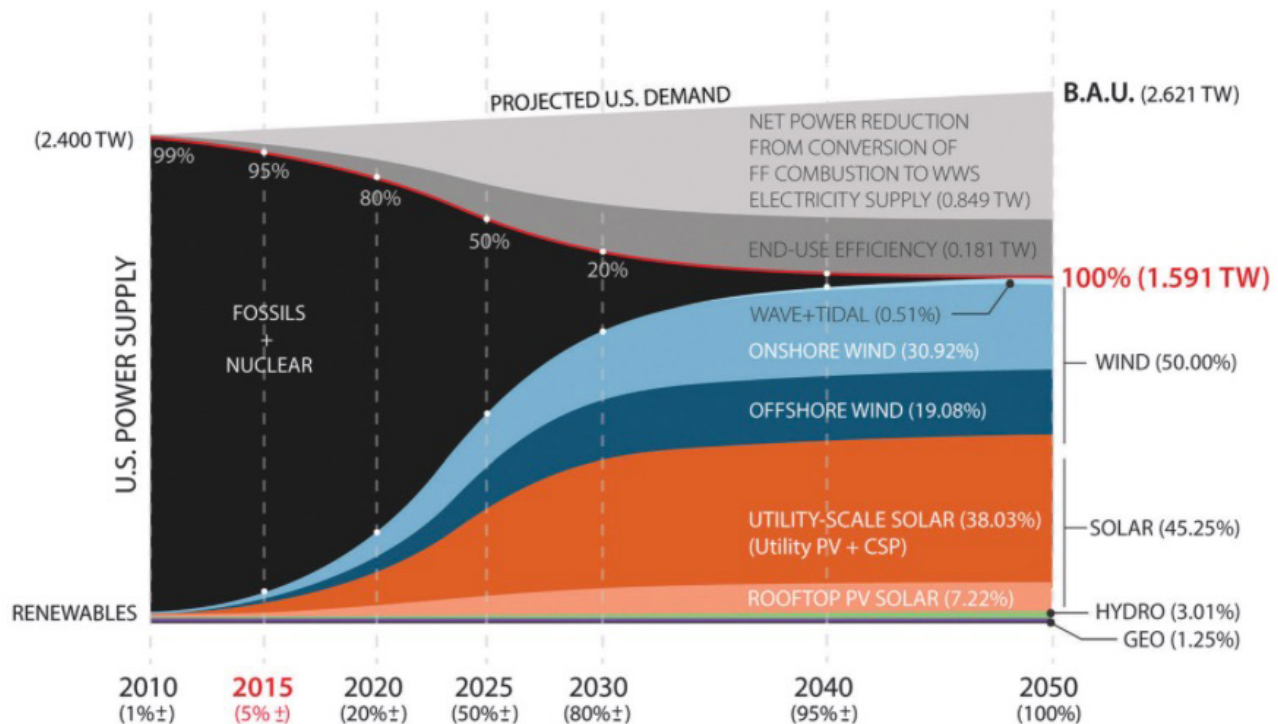


Technological advancement is no longer preventing the U.S. from powering the nation with renewable energy, finds a new study. Many other forces: political, economic and public interest are all at various stages of opposition, but a recent study finds that if all agree to pursue a clean energy future, the U.S. would be able to operate solely on renewables by 2050.

Mark Z. Jacobson, a civil and environmental engineer at Stanford's Atmosphere and Energy Program authored the study. The findings were first published (<http://rsc.li/1LQvZuY>) in Energy & Environmental Science, a publication that links "all aspects of the chemical, physical and biotechnological sciences relating to energy conversion and storage, alternative fuel technologies and environmental science."

The figure of 100% renewable energy for the U.S. is an audacious goal which depends on a ground swell of support. But, the idea of shifting so quickly and fully to clean energy in this country is not that farfetched. The growth of solar in the U.S. continues to exceed expectations. Projections regularly need tweaking as benchmarks are hit early.

The study presents an exponential increase that is already underway will continue and accelerate. “The plans contemplate 80%–85% of existing energy replaced by 2030 and 100% replaced by 2050,” says (<http://rsc.li/1LQvZuY>) Jacobson. He acknowledges that overall energy use will decrease over time because of continued advancements in energy efficiency. The study includes a simple graph to explain a rather complex energy projection:



Source: Jacobson et al., *Energy & Environmental Science*, 2015

The chart shows the U.S. will likely decrease yearly energy demand by nearly a terawatt, from 2,400 TW annually down to 1,591 TW by 2050. This will be achieved partially by a net power reduction from conversions related to dropping fossil fuels for wind and solar generated electricity, as well as an increased efficiency of end user operations.

Interestingly, efficiencies from power generation will likely eclipse the savings from end user efficiency increases, which have already seen a dramatic shift towards energy conservation. We've seen items like LED light bulbs and flat panel TVs prove a huge end user decrease in power consumption in just the past decade alone. This study assumes a rapid increase in gains along those lines.

Devil in the Details

While it's easy to get swept up in the overall positive prediction, it is important to understand what will be required from an energy production standpoint.

The study suggests that wind and solar generating power plants have a decreased percentage of time generating a maximum output of power. “To produce that much energy, they propose building power plants with a total of 6.5 TW of capacity. By way of comparison, the US currently has about 1.2 TW of installed electric generation capacity, so this plan would involve expanding generation capacity fivefold in 35 years,” reports (<http://bit.ly/1HbwUjp>) David Roberts at Vox.

Of course this audacious goal requires, as the study puts it, “strong enough political will.” Jacobson cites the nation's ramp up of aircraft production during World War II from “nearly zero to 330,000 over five years.”

Unfortunately, as history has shown, political interest in rapid development during war time is far higher than when discussing an upheaval of the entrenched fossil fuel energy production industry and replacing it with clean energy.

Takeaway

Lofty aspirations for a prosperous and clean future for all Americans could be seen as a noble goal. But the realities of the political climate in this country may keep the U.S. from hitting the audacious goals outlined in this study. This doesn't prevent homeowners from using the same technology outlined by Jacobson for their own home use. The benefit of the rapidly advancing solar panel technology is that each homeowner can become an independent power generator. And that may be a far more realistic and achievable goal by 2050.

Availability of any state, federal, or local programs will vary and are subject to change. Consult with a financial professional before making any decisions. Energy and financial savings are not guaranteed and will vary from customer to customer. The foregoing is provided for informational purposes only and is not expert or professional advice, and should not be relied upon as such. SolarMeHome does not make any recommendations for any particular solar installer or product. SolarMeHome receives payment from participating installers for customer referrals.

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